

Fundamentals of Mathematics (25VBT0C102)



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Module: 04

Integral Calculus & Vector Calculus

Module Information

Field	Details
Course Title	Fundamentals of Mathematics
Course Code	25VBT0C102
Module Number	Module 4 of 5
Module Title	Integral Calculus and Vector Calculus

Module Overview

Integration is the second great operation of calculus — the companion to differentiation studied in Module 3. Where differentiation asks "at what rate is this quantity changing?", integration asks "what is the total accumulation of this rate of change?" These two operations are inverses of each other, a profound relationship captured in the Fundamental Theorem of Calculus. Together, differentiation and integration form the complete language of continuous change.

The first half of this module develops the full toolkit of integration techniques. Indefinite integration — finding the antiderivative — begins with the reverse of the standard differentiation rules, then expands through four powerful methods: integration by substitution (the reverse of the chain rule), integration using trigonometric identities (for powers and products of trigonometric functions), the method of partial fractions (for rational functions), and integration by parts (the reverse of the product rule). Each technique is suited to different classes of integrands, and knowing which technique to apply is a key analytical skill.

Definite integrals measure the net area under a curve between two limits. The Fundamental Theorem of Calculus connects the definite integral to the antiderivative, giving a practical computation method. Special attention is given to reduction formulae — recursive formulas that express the integral of $\sin^n x$ or $\cos^n x$ in terms of an integral of a lower power. These formulae are elegant, efficient, and appear repeatedly in physics, statistics, and engineering applications.

The second half of the module transitions to vector calculus — the extension of calculus to vector-valued functions and vector fields. Vector functions describe quantities that vary in both magnitude and direction through space: the velocity and acceleration of a particle, the flow of a fluid, or the electric field around a charged object. The derivative of a vector function gives the instantaneous velocity vector; the second derivative gives acceleration.

Vector calculus introduces three fundamental differential operators that act on fields: the gradient (∇f), which points in the direction of steepest increase of a scalar field; the divergence ($\nabla \cdot \mathbf{F}$), which measures the net outflow of a vector field at each point; and the curl ($\nabla \times \mathbf{F}$), which measures the rotational tendency of a vector field. The Laplacian ($\nabla^2 f = \nabla \cdot \nabla f$) combines gradient and divergence and appears in the fundamental equations of heat, wave, and electrostatic theory.

Two important classes of vector fields are studied: solenoidal (divergence-free) fields, which represent incompressible fluid flow and magnetic fields, and irrotational (curl-free) fields, which represent conservative force fields such as gravity and electrostatics. Understanding these special properties of vector fields is essential for applications in fluid mechanics, electromagnetic theory, and computational physics.

Throughout the module, the emphasis is on clean computation, geometric understanding, and real-world application. The mathematical tools developed here are the foundation of Maxwell's equations of electromagnetism, the Navier-Stokes equations of fluid mechanics, and the heat and Schrödinger equations of physics — arguably the most important equations in all of science.

Learning Objectives

By the end of this module, you will be able to:

- Apply standard integration techniques — substitution, trigonometric identities, partial fractions, and integration by parts — to evaluate indefinite integrals of algebraic, trigonometric, exponential, and rational functions.
- Evaluate definite integrals using the Fundamental Theorem of Calculus and apply reduction formulae for powers of sine and cosine.
- Understand vector-valued functions and compute their derivatives to find velocity and acceleration vectors.
- Understand scalar and vector point functions and apply the differential operators — gradient, divergence, curl, and Laplacian — to analyse physical fields.
- Identify solenoidal and irrotational vector fields and interpret their physical significance in fluid mechanics and electromagnetism.

Learning Outcomes

After completing this module, you will be able to:

- Select and apply the appropriate integration method for a given integral and evaluate it correctly, including definite integrals with limits.
- Apply reduction formulae to evaluate $\int \sin^n x \, dx$, $\int \cos^n x \, dx$, and related integrals over standard limits.
- Differentiate vector functions to compute velocity, acceleration, and unit tangent vectors.
- Compute the gradient of scalar fields, divergence and curl of vector fields, and the Laplacian of scalar functions.
- Test whether a vector field is solenoidal ($\nabla \cdot \mathbf{F} = 0$) or irrotational ($\nabla \times \mathbf{F} = 0$) and interpret the physical meaning.

Table of Topics

Sub-Unit	Topic	Description
4.1	Integration — Definition and Standard Results	Antiderivative, indefinite integral, constant of integration, table of standard forms
4.2	Integration by Substitution	Reversing the chain rule; choosing substitution $u = g(x)$
4.3	Integration Using Trigonometric Identities	Powers of $\sin/\cos$, products, half-angle and double-angle identities
4.4	Integration by Partial Fractions	Rational functions, proper/improper fractions, linear/quadratic/repeated factors
4.5	Integration by Parts	$\int u \, dv = uv - \int v \, du$; ILATE rule; repeated application
4.6	Definite Integrals and the Fundamental Theorem	Limits of integration, area interpretation, properties of definite integrals
4.7	Reduction Formulae	I_n for $\sin^n x$ and $\cos^n x$; Wallis formula; recursive derivation
4.8	Vector Functions and Derivatives	$\mathbf{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$; $d\mathbf{r}/dt$; velocity, speed, acceleration
4.9	Scalar and Vector Point Functions	Fields in 3D; scalar field $\phi(x,y,z)$ vs. vector field $\mathbf{F}(x,y,z)$
4.10	Gradient of a Scalar Field	$\nabla\phi = (\partial\phi/\partial x)\hat{i} + (\partial\phi/\partial y)\hat{j} + (\partial\phi/\partial z)\hat{k}$; directional derivative
4.11	Divergence of a Vector Field	$\nabla \cdot \mathbf{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$; physical interpretation as source/sink
4.12	Curl of a Vector Field	$\nabla \times \mathbf{F}$ — rotational measure; determinant form; irrotational fields
4.13	Laplacian Operator	$\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$; harmonic functions
4.14	Solenoidal and Irrotational Vector Fields	$\nabla \cdot \mathbf{F} = 0$ (incompressible); $\nabla \times \mathbf{F} = 0$ (conservative); physical examples

4.1 Integration — Definition and Standard Results

Integration, in its most fundamental form, is the reverse process of differentiation. Given a function $f(x)$, we seek a function $F(x)$ whose derivative is $f(x)$ — such a function is called an antiderivative or primitive of $f(x)$. The process of finding it is called integration.

The indefinite integral of $f(x)$ with respect to x , written $\int f(x) dx$, represents the family of all antiderivatives of $f(x)$. Since the derivative of any constant is zero, the antiderivative is determined only up to an arbitrary constant C — called the constant of integration:

Indefinite integral: $\int f(x) dx = F(x) + C$ where $F'(x) = f(x)$

The symbol $\int$ (an elongated S , from Latin "summa") and the dx both come from Leibniz, reminding us that integration originated as summation of infinitely thin strips. The integrand $f(x)$ is the function being integrated, and dx indicates the variable of integration.

Fundamental Theorem of Calculus (Preview)

Differentiation and integration are inverse operations. If $F'(x) = f(x)$, then $\int f(x) dx = F(x) + C$. Conversely, $d/dx[\int f(x) dx] = f(x)$. This inverse relationship is the cornerstone of the entire subject.

Table of Standard Integration Results

Integrand $f(x)$	Integral $\int f(x) dx$ (+ C implied)
x^n ($n \neq -1$)	$x^{n+1}/(n+1)$
$1/x$ ($x > 0$)	$\ln x $
e^x	e^x
a^x ($a > 0$)	$a^x/\ln a$
$\sin x$	$-\cos x$
$\cos x$	$\sin x$
$\sec^2 x$	$\tan x$
$\operatorname{cosec}^2 x$	$-\cot x$
$\sec x \cdot \tan x$	$\sec x$
$\operatorname{cosec} x \cdot \cot x$	$-\operatorname{cosec} x$
$1/\sqrt{1-x^2}$	$\sin^{-1}x$ (or $-\cos^{-1}x$)
$1/(1+x^2)$	$\tan^{-1}x$
$1/(a^2+x^2)$	$(1/a) \cdot \tan^{-1}(x/a)$
$1/\sqrt{a^2-x^2}$	$\sin^{-1}(x/a)$
$e^x[f(x) + f'(x)]$	$e^x \cdot f(x)$ — the e^x shortcut rule

★ **Note:** Every integration result can be verified by differentiation: if $\int f(x) dx = F(x) + C$, then $F'(x)$ must equal $f(x)$. This provides a built-in check for every integral you compute.

4.2 Integration by Substitution

Integration by substitution is the most frequently used integration technique. It is the counterpart of the chain rule in differentiation. The idea is to replace a complex expression in the integrand with a single new variable u , simplify the integral, evaluate it in terms of u , and then substitute back.

If the integrand has the form $f(g(x)) \cdot g'(x)$, we substitute $u = g(x)$, giving $du = g'(x)dx$, and the integral becomes $\int f(u) du$ — often much simpler. The key skill is recognising the structure: one part of the integrand should be the derivative (or a multiple of the derivative) of another part.

Substitution formula: $\int f(g(x)) \cdot g'(x) dx = \int f(u) du$ where $u = g(x)$, $du = g'(x)dx$

Standard Substitution Patterns

- Pattern 1 — Linear: If integrand contains $(ax+b)^n$, substitute $u = ax+b$.
- Pattern 2 — Chain: If integrand has $f(g(x)) \cdot g'(x)$, substitute $u = g(x)$.
- Pattern 3 — Trigonometric: For $\sqrt{a^2-x^2}$, use $x = a \sin \theta$; for $\sqrt{a^2+x^2}$, use $x = a \tan \theta$; for $\sqrt{x^2-a^2}$, use $x = a \sec \theta$.

Worked Example: Integration by Substitution — Standard Cases

Evaluate: (i) $\int 2x \cdot e^{(x^2)} dx$ (ii) $\int \sin^5 x \cdot \cos x dx$ (iii) $\int dx / \sqrt{4-x^2}$

Step 1: (i) $\int 2x \cdot e^{(x^2)} dx$. Let $u = x^2$, then $du = 2x dx$.

$$\int 2x \cdot e^{(x^2)} dx = \int e^u du = e^u + C = e^{(x^2)} + C$$

Step 2: (ii) $\int \sin^5 x \cdot \cos x dx$. Let $u = \sin x$, $du = \cos x dx$.

$$\int \sin^5 x \cdot \cos x dx = \int u^5 du = u^6/6 + C = \sin^6 x/6 + C$$

Step 3: (iii) $\int dx / \sqrt{4-x^2}$. Let $x = 2 \sin \theta$, $dx = 2 \cos \theta d\theta$, $\sqrt{4-x^2} = 2 \cos \theta$.

$$\int (2 \cos \theta d\theta) / (2 \cos \theta) = \int d\theta = \theta + C = \sin^{-1}(x/2) + C$$

** Worked Example: Substitution with Partial Adjustment**

Evaluate: (i) $\int x/(x^2+1) dx$ (ii) $\int (3x+1)^4 dx$

Step 1: (i) Let $u = x^2+1$, $du = 2x dx \Rightarrow x dx = du/2$

$$\int x/(x^2+1) dx = \int (1/u)(du/2) = (1/2)\ln|u| + C = (1/2)\ln(x^2+1) + C$$

Step 2: (ii) Let $u = 3x+1$, $du = 3 dx \Rightarrow dx = du/3$

$$\int (3x+1)^4 dx = \int u^4(du/3) = (1/3) \cdot u^5/5 + C = (3x+1)^5/15 + C$$

4.3 Integration Using Trigonometric Identities

Many integrals involving powers and products of trigonometric functions resist direct substitution but yield elegantly to trigonometric identities. The strategy is to use standard identities to rewrite the integrand in a simpler form — typically as a sum of simpler trigonometric functions — before integrating term by term.

Key Identities Used in Integration

- Pythagorean: $\sin^2 x + \cos^2 x = 1$; also $\sin^2 x = 1 - \cos^2 x$ and $\cos^2 x = 1 - \sin^2 x$
- Half-angle (power reduction): $\sin^2 x = (1 - \cos 2x)/2$ and $\cos^2 x = (1 + \cos 2x)/2$
- Double-angle: $\sin 2x = 2 \sin x \cos x$ and $\cos 2x = \cos^2 x - \sin^2 x$
- Product-to-sum: $\sin A \cos B = (1/2)[\sin(A+B) + \sin(A-B)]$
- Sec-tan: $\sec^2 x = 1 + \tan^2 x$ and $\operatorname{cosec}^2 x = 1 + \cot^2 x$

Worked Example: Integrals of Powers of sin and cos

Evaluate: (i) $\int \sin^2 x \, dx$ (ii) $\int \cos^4 x \, dx$ (iii) $\int \sin^3 x \, dx$

Step 1: (i) $\int \sin^2 x \, dx$. Use half-angle: $\sin^2 x = (1 - \cos 2x)/2$.

$$\int \sin^2 x \, dx = \int (1 - \cos 2x)/2 \, dx = x/2 - \sin(2x)/4 + C$$

Step 2: (ii) $\int \cos^4 x \, dx$. Write $\cos^4 x = (\cos^2 x)^2 = [(1 + \cos 2x)/2]^2 = (1 + 2\cos 2x + \cos^2 2x)/4$.

Use $\cos^2 2x = (1 + \cos 4x)/2$:

$$\cos^4 x = (1 + 2\cos 2x + (1 + \cos 4x)/2)/4 = (3/8) + (1/2)\cos 2x + (1/8)\cos 4x$$

$$\int \cos^4 x \, dx = 3x/8 + \sin(2x)/4 + \sin(4x)/32 + C$$

Step 3: (iii) $\int \sin^3 x \, dx$. Write $\sin^3 x = \sin x(1 - \cos^2 x) = \sin x - \sin x \cos^2 x$.

For second term, let $u = \cos x$, $du = -\sin x \, dx$:

$$\int \sin^3 x \, dx = -\cos x + \cos^3 x/3 + C$$

Worked Example: Product of sin and cos

Evaluate: $\int \sin 3x \cdot \cos 2x \, dx$

Step 1: Use product-to-sum: $\sin A \cos B = (1/2)[\sin(A+B) + \sin(A-B)]$

$$\sin 3x \cdot \cos 2x = (1/2)[\sin(5x) + \sin(x)]$$

Step 2: $\int \sin 3x \cos 2x \, dx = (1/2) \int [\sin 5x + \sin x] \, dx$

$$= (1/2)[- \cos(5x)/5 - \cos x] + C = -\cos(5x)/10 - \cos(x)/2 + C$$

 Let's Check

Q.No	Type	Question
1	MCQ	$\int e^x dx$ equals:
2	MCQ	Using substitution $u = x^2$, $\int 2x \cos(x^2) dx$ equals:
3	Fill in Blank	$\int \sec^2 x dx = \underline{\hspace{2cm}}$.
4	Fill in Blank	The substitution for $\int dx/\sqrt{a^2-x^2}$ is $x = \underline{\hspace{2cm}}$.
5	True/False	$\int \sin^2 x dx = -\sin^2 x/2 + C$.
6	True/False	Integration by substitution is the reverse of the chain rule of differentiation.

Answers:

Q.No	Answer
1	b) $e^x + C$ — the exponential function is its own antiderivative.
2	c) $\sin(x^2) + C$ — after substitution: $\int \cos(u) du = \sin u + C = \sin(x^2) + C$.
3	$\tan x + C$
4	$a \sin \theta$
5	False — $\int \sin^2 x dx = x/2 - \sin(2x)/4 + C$, using the identity $\sin^2 x = (1 - \cos 2x)/2$.
6	True — if $u = g(x)$, then $\int f(g(x))g'(x)dx = \int f(u)du$, reversing the chain rule.

4.4 Integration by Partial Fractions

The method of partial fractions decomposes a proper rational function — a ratio of polynomials where the degree of the numerator is less than the degree of the denominator — into a sum of simpler fractions, each of which can be integrated using standard results. This technique is essential for integrating rational functions, which appear in engineering, control systems, and signal processing.

If the rational function is improper (degree of numerator $\geq$ degree of denominator), first perform polynomial long division to obtain a polynomial plus a proper rational function. Then apply partial fractions to the proper part.

Types of Factors and Their Partial Fraction Forms

Factor Type in Denominator	Partial Fraction Form	Notes
Non-repeated linear: $(x-a)$	$A/(x-a)$	Numerator A is a constant to be found
Repeated linear: $(x-a)^2$	$A/(x-a) + B/(x-a)^2$	Two constants A, B needed
Repeated linear: $(x-a)^n$	$A_1/(x-a) + A_2/(x-a)^2 + \dots + A_n/(x-a)^n$	n constants needed
Irreducible quadratic: (x^2+bx+c)	$(Ax+B)/(x^2+bx+c)$	Linear numerator (A, B constants)
Repeated quadratic: $(x^2+bx+c)^2$	$(Ax+B)/(x^2+bx+c) + (Cx+D)/(x^2+bx+c)^2$	Two linear numerators

Worked Example: Partial Fractions — Distinct Linear Factors

Evaluate: $\int (3x+2)/[(x+1)(x+2)] dx$

Step 1: Decompose: $(3x+2)/[(x+1)(x+2)] = A/(x+1) + B/(x+2)$

Multiply both sides by $(x+1)(x+2)$: $3x+2 = A(x+2) + B(x+1)$

Step 2: Find A: let $x = -1$: $3(-1)+2 = A(1) \Rightarrow -1 = A \Rightarrow A = -1$

Step 3: Find B: let $x = -2$: $3(-2)+2 = B(-1) \Rightarrow -4 = -B \Rightarrow B = 4$

Step 4: Integrate:

$$\begin{aligned}\int (3x+2)/[(x+1)(x+2)] dx &= \int [-1/(x+1) + 4/(x+2)] dx \\ &= -\ln|x+1| + 4 \ln|x+2| + C\end{aligned}$$

Worked Example: Partial Fractions — Repeated Factor

Evaluate: $\int (x+3)/[(x-1)^2(x+1)] dx$

Step 1: Decompose: $(x+3)/[(x-1)^2(x+1)] = A/(x-1) + B/(x-1)^2 + C/(x+1)$

Multiply out: $x+3 = A(x-1)(x+1) + B(x+1) + C(x-1)^2$

Step 2: $x = 1$: $4 = B(2) \Rightarrow B = 2$

Step 3: $x = -1$: $2 = C(4) \Rightarrow C = 1/2$

Step 4: $x = 0$: $3 = A(-1)(1) + 2(1) + (1/2)(1) \Rightarrow 3 = -A + 5/2 \Rightarrow A = -1/2$

Step 5: Integrate:

$$\begin{aligned}\int &= (-1/2)\ln|x-1| + 2 \cdot (-1/(x-1)) + (1/2)\ln|x+1| + C \\ &= (1/2)\ln|(x+1)/(x-1)| - 2/(x-1) + C\end{aligned}$$

4.5 Integration by Parts

Integration by parts is the integration counterpart of the product rule of differentiation. It is used when the integrand is a product of two functions from different families — for example, a polynomial times an exponential, or a polynomial times a trigonometric function. The formula transforms one integral into another that is (hopefully) easier to evaluate.

If u and v are both differentiable functions of x , then:

Integration by Parts:
$$\int u \, dv = uv - \int v \, du$$

Equivalently, for two functions $f(x)$ and $g(x)$:

Product form:
$$\int f(x) \cdot g'(x) \, dx = f(x) \cdot g(x) - \int g(x) \cdot f'(x) \, dx$$

The ILATE Rule — Choosing u

The key to applying integration by parts is choosing which function to call u (to differentiate) and which to call dv (to integrate). The ILATE priority rule gives the conventional order of preference for u :

- I — Inverse trigonometric functions: $\sin^{-1}x$, $\tan^{-1}x$, etc.
- L — Logarithmic functions: $\ln x$, $\log x$
- A — Algebraic functions: polynomials, x^n
- T — Trigonometric functions: $\sin x$, $\cos x$, etc.
- E — Exponential functions: e^x , a^x

Choose u as the function that appears first in ILATE, and dv as the rest of the integrand. This ordering minimises the complexity of $\int v \, du$.

Worked Example: Integration by Parts — Standard Cases

Evaluate: (i) $\int x \cdot e^x dx$ (ii) $\int x \cdot \sin x dx$ (iii) $\int \ln x dx$

Step 1: (i) $\int x \cdot e^x dx$. By ILATE: $u = x$ (A before E), $dv = e^x dx$.

$$du = dx, \quad v = e^x$$

$$\int x \cdot e^x dx = x \cdot e^x - \int e^x dx = x \cdot e^x - e^x + C = e^x(x-1) + C$$

Step 2: (ii) $\int x \cdot \sin x dx$. By ILATE: $u = x$ (A before T), $dv = \sin x dx$.

$$du = dx, \quad v = -\cos x$$

$$\int x \cdot \sin x dx = -x \cdot \cos x - \int (-\cos x) dx = -x \cos x + \sin x + C$$

Step 3: (iii) $\int \ln x dx$. Write as $\int (\ln x) \cdot 1 dx$. By ILATE: $u = \ln x$ (L before A), $dv = dx$.

$$du = (1/x)dx, \quad v = x$$

$$\int \ln x dx = x \cdot \ln x - \int x \cdot (1/x) dx = x \cdot \ln x - \int 1 dx = x \cdot \ln x - x + C = x(\ln x - 1) + C$$

Worked Example: Repeated Integration by Parts

Evaluate $\int x^2 \cdot e^x dx$.

Step 1: First application: $u = x^2$, $dv = e^x dx$. $du = 2x dx$, $v = e^x$.

$$\int x^2 \cdot e^x dx = x^2 \cdot e^x - \int e^x \cdot 2x dx = x^2 e^x - 2 \int x \cdot e^x dx$$

Step 2: Second application for $\int x \cdot e^x dx$ (from worked example above): $= e^x(x-1)$

$$\int x^2 \cdot e^x dx = x^2 e^x - 2e^x(x-1) + C = e^x(x^2 - 2x + 2) + C$$

🔍 Analysing Misconceptions: Integration is Simply "Reverse Differentiation"

✗ MYTH: Every function has a nice antiderivative expressible in elementary functions, just as every function has a derivative.

✓ REALITY: This is false. While every continuous function theoretically has an antiderivative, many simple-looking functions have no antiderivative expressible in terms of elementary functions (polynomials, exponentials, trig functions, logarithms, and their inverses). Famous examples include: $\int e^{-x^2} dx$ (the Gaussian integral, crucial in statistics), $\int \sin(x)/x dx$, $\int 1/\ln(x) dx$, and $\int \sqrt{1 - k^2 \sin^2 x} dx$ (elliptic integral). These integrals are evaluated numerically or using special functions, not by algebraic antiderivatives.

4.6 Definite Integrals and the Fundamental Theorem of Calculus

The definite integral extends the concept of an antiderivative to compute a specific numerical value — the net area under a curve between two bounds. This is a profound concept: we compute the cumulative effect of a continuously varying quantity. Applications include the area under a demand curve, total distance from a velocity function, work done by a variable force, and probability from a density function.

The Fundamental Theorem of Calculus

The Fundamental Theorem connects the two halves of calculus — differentiation and integration — in one of the most important theorems in mathematics. It has two parts:

- Part 1 — Differentiation of an integral: If $F(x) = \int [a \text{ to } x] f(t) dt$, then $F'(x) = f(x)$.
An integral with a variable upper limit is an antiderivative of the integrand.
- Part 2 — Evaluation theorem: If F is any antiderivative of f on $[a, b]$ (i.e., $F'(x) = f(x)$), then:

Definite integral evaluation: $\int [a \text{ to } b] f(x) dx = F(b) - F(a) = [F(x)]_a^b$

Properties of Definite Integrals

- $\int [a \text{ to } b] f(x) dx = -\int [b \text{ to } a] f(x) dx$ — reversing limits negates the integral.
- $\int [a \text{ to } a] f(x) dx = 0$ — integral over a point is zero.
- $\int [a \text{ to } b] [f(x) \pm g(x)] dx = \int [a \text{ to } b] f(x) dx \pm \int [a \text{ to } b] g(x) dx$ — linearity.
- $\int [a \text{ to } b] f(x) dx = \int [a \text{ to } c] f(x) dx + \int [c \text{ to } b] f(x) dx$ — additivity over intervals.
- If f is even [$f(-x) = f(x)$]: $\int [-a \text{ to } a] f(x) dx = 2 \int [0 \text{ to } a] f(x) dx$.
- If f is odd [$f(-x) = -f(x)$]: $\int [-a \text{ to } a] f(x) dx = 0$.
- King's property: $\int [0 \text{ to } a] f(x) dx = \int [0 \text{ to } a] f(a-x) dx$ — very useful for trigonometric definite integrals.

Worked Example: Evaluating Definite Integrals

Evaluate: (i) $\int [0 \text{ to } \pi/2] \sin x \, dx$ (ii) $\int [1 \text{ to } 2] (3x^2 - 2x + 1) \, dx$ (iii) $\int [0 \text{ to } 1] x/(x^2+1) \, dx$

Step 1: (i) $\int [0 \text{ to } \pi/2] \sin x \, dx = [-\cos x]_0^{\pi/2} = (-\cos(\pi/2)) - (-\cos 0) = 0 - (-1) = 1$

Step 2: (ii) $\int [1 \text{ to } 2] (3x^2 - 2x + 1) \, dx = [x^3 - x^2 + x]_1^2 = (8 - 4 + 2) - (1 - 1 + 1) = 6 - 1 = 5$

Step 3: (iii) $\int [0 \text{ to } 1] x/(x^2+1) \, dx$. Let $u = x^2+1$, $du = 2x \, dx$. When $x=0$: $u=1$; $x=1$: $u=2$.
 $= \int [1 \text{ to } 2] (1/u)(du/2) = (1/2)[\ln|u|]_1^2 = (1/2)(\ln 2 - \ln 1) = (\ln 2)/2$

4.7 Reduction Formulae for Powers of Sine and Cosine

Reduction formulae are recursive integral formulas that express the integral of $\sin^n x$ or $\cos^n x$ in terms of the integral of a lower power of the same function. They avoid having to apply repeated integration by parts for each new power, instead building systematically from lower to higher (or higher to lower) powers.

Reduction Formula for $\sin^n x$

Using integration by parts with $u = \sin^{n-1} x$ and $dv = \sin x \, dx$, one derives:

Reduction formula for $\sin^n x$: $\int \sin^n x \, dx = -(\sin^{n-1} x \cdot \cos x)/n + (n-1)/n \cdot \int \sin^{n-2} x \, dx$

Reduction Formula for $\cos^n x$

By a symmetric derivation using $u = \cos^{n-1} x$ and $dv = \cos x \, dx$:

Reduction formula for $\cos^n x$: $\int \cos^n x \, dx = (\cos^{n-1} x \cdot \sin x)/n + (n-1)/n \cdot \int \cos^{n-2} x \, dx$

Wallis Formula — Definite Integral from 0 to $\pi/2$

When evaluated between 0 and $\pi/2$, the reduction formulae give the elegantly simple Wallis formula. Denote $I_n = \int [0 \text{ to } \pi/2] \sin^n x \, dx = \int [0 \text{ to } \pi/2] \cos^n x \, dx$ (they are equal by symmetry). Then:

Wallis reduction: $I_n = [(n-1)/n] \cdot I_{n-2}$

Starting values: $I_0 = \pi/2$ and $I_1 = 1$. The pattern gives:

- n even: $I_n = [(n-1)/n] \cdot [(n-3)/(n-2)] \cdots [1/2] \cdot (\pi/2)$
- n odd: $I_n = [(n-1)/n] \cdot [(n-3)/(n-2)] \cdots [2/3] \cdot 1$

Worked Example: Applying Reduction Formulae — Wallis

Evaluate: (i) $\int [0 \text{ to } \pi/2] \sin^4 x \, dx$ (ii) $\int [0 \text{ to } \pi/2] \cos^5 x \, dx$

Step 1: (i) $I_4 = (3/4) \cdot I_2 = (3/4) \cdot (1/2) \cdot I_0 = (3/4) \cdot (1/2) \cdot (\pi/2) = 3\pi/16$

Step 2: Verify by direct computation: $\int [0 \text{ to } \pi/2] \sin^4 x \, dx$

$$\sin^4 x = (3/8) - (1/2)\cos 2x + (1/8)\cos 4x \text{ (from identity in 4.3)}$$

$$= [3x/8 - \sin(2x)/4 + \sin(4x)/32]_0^{\pi/2} = 3\pi/16 - 0 + 0 = 3\pi/16 \quad \checkmark$$

Step 3: (ii) $I_5 = (4/5) \cdot I_3 = (4/5) \cdot (2/3) \cdot I_1 = (4/5) \cdot (2/3) \cdot 1 = 8/15$

Worked Example: Reduction Formula — Indefinite Integral $\sin^4 x$

Use the reduction formula to find $\int \sin^4 x \, dx$.

Step 1: Apply $n=4$: $\int \sin^4 x \, dx = -(\sin^3 x \cdot \cos x)/4 + (3/4) \int \sin^2 x \, dx$

Step 2: Apply $n=2$: $\int \sin^2 x \, dx = -(\sin x \cdot \cos x)/2 + (1/2) \int \sin^0 x \, dx = -(\sin x \cdot \cos x)/2 + x/2$

Step 3: Combine:

$$\begin{aligned} \int \sin^4 x \, dx &= -(\sin^3 x \cdot \cos x)/4 + (3/4)[x/2 - (\sin x \cdot \cos x)/2] + C \\ &= -(\sin^3 x \cdot \cos x)/4 + 3x/8 - (3 \sin x \cos x)/8 + C \end{aligned}$$

Using $\sin 2x = 2 \sin x \cos x$: $= 3x/8 - (\sin 2x)/4 - (\sin^3 x \cdot \cos x)/4 + C$

✓ Let's Check

Q.No	Type	Question
1	MCQ	$\int \ln x \, dx$ equals:
2	MCQ	For $\int [0 \text{ to } \pi/2] \sin^n x \, dx$, the Wallis formula gives I_6 as:
3	Fill in Blank	In integration by parts, the ILATE rule helps choose the function _____ to differentiate.
4	Fill in Blank	$\int [0 \text{ to } \pi/2] \sin^5 x \, dx = (4/5) \cdot (2/3) \cdot \underline{\hspace{2cm}}$ using the Wallis formula.
5	True/False	$\int [a \text{ to } b] f(x) dx = \int [0 \text{ to } a] f(a-x) dx$ is always true for any a .
6	True/False	For an odd function $f(x)$, $\int [-a \text{ to } a] f(x) dx = 0$.

Answers:

Q.No	Answer
1	b) $x(\ln x - 1) + C$ — use integration by parts with $u = \ln x$, $dv = dx$.
2	c) $5\pi/32$ — $I_6 = (5/6) \cdot (3/4) \cdot (1/2) \cdot (\pi/2) = 5\pi/32$.
3	u (the first function, to be differentiated)
4	$I_1 = 1 \rightarrow I_5 = 8/15$
5	False — The correct king's property is: $\int [0 \text{ to } a] f(x) dx = \int [0 \text{ to } a] f(a-x) dx$.

6

True — for odd f , the negative area on $[-a,0]$ exactly cancels the positive area on $[0,a]$.

4.8 Vector Functions and Their Derivatives

The second major topic of this module is vector calculus — the calculus of functions that have vector values or take vector arguments. The simplest setting is a vector-valued function of a scalar parameter, usually time t . Such a function $\mathbf{r}(t)$ describes the position of a particle at time t , tracing a path through three-dimensional space.

A vector function $\mathbf{r}(t)$ assigns a vector to each value of t . In three dimensions:

Vector function: $\mathbf{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k} = (f(t), g(t), h(t))$

where f, g, h are ordinary scalar functions of t called the component functions.

Derivative of a Vector Function

The derivative of $\mathbf{r}(t)$ is defined componentwise — exactly as the limit of the difference quotient, but applied to each component independently:

Derivative of vector function: $\mathbf{r}'(t) = d\mathbf{r}/dt = f'(t)\hat{i} + g'(t)\hat{j} + h'(t)\hat{k}$

This is simply: differentiate each component. The derivative vector $\mathbf{r}'(t)$ points in the direction of the instantaneous tangent to the curve at the point $\mathbf{r}(t)$ — it is the velocity vector.

Velocity, Speed, and Acceleration

- Velocity vector: $\mathbf{v}(t) = \mathbf{r}'(t) = (dx/dt)\hat{i} + (dy/dt)\hat{j} + (dz/dt)\hat{k}$
- Speed (scalar): $|\mathbf{v}(t)| = \sqrt{(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2}$
- Acceleration vector: $\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t) = (d^2x/dt^2)\hat{i} + (d^2y/dt^2)\hat{j} + (d^2z/dt^2)\hat{k}$
- Unit tangent vector: $\hat{T}(t) = \mathbf{v}(t)/|\mathbf{v}(t)|$ — unit vector in direction of motion

Worked Example: Velocity and Acceleration from Vector Function

A particle moves with $\mathbf{r}(t) = (3t^2)\hat{i} + (2t^3 - t)\hat{j} + (e^t)\hat{k}$. Find velocity, speed, and acceleration at $t = 0$.

Step 1: Velocity: $\mathbf{v}(t) = \mathbf{r}'(t) = 6t\hat{i} + (6t^2 - 1)\hat{j} + e^t\hat{k}$

Step 2: At $t = 0$: $\mathbf{v}(0) = 0\hat{i} + (-1)\hat{j} + 1\hat{k} = -\hat{j} + \hat{k}$

Step 3: Speed at $t = 0$: $|\mathbf{v}(0)| = \sqrt{0 + 1 + 1} = \sqrt{2}$

Step 4: Acceleration: $\mathbf{a}(t) = \mathbf{v}'(t) = 6\hat{i} + 12t\hat{j} + e^t\hat{k}$

Step 5: At $t = 0$: $\mathbf{a}(0) = 6\hat{i} + 0\hat{j} + \hat{k} = 6\hat{i} + \hat{k}$

Differentiation Rules for Vector Functions

Vector functions satisfy differentiation rules analogous to scalar calculus:

- $\frac{d}{dt}[\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \mathbf{r}_1'(t) + \mathbf{r}_2'(t)$ — sum rule
- $\frac{d}{dt}[\mathbf{f}(t) \cdot \mathbf{r}(t)] = \mathbf{f}'(t) \cdot \mathbf{r}(t) + \mathbf{f}(t) \cdot \mathbf{r}'(t)$ — scalar product rule
- $\frac{d}{dt}[\mathbf{r}_1 \cdot \mathbf{r}_2] = \mathbf{r}_1' \cdot \mathbf{r}_2 + \mathbf{r}_1 \cdot \mathbf{r}_2'$ — dot product rule
- $\frac{d}{dt}[\mathbf{r}_1 \times \mathbf{r}_2] = \mathbf{r}_1' \times \mathbf{r}_2 + \mathbf{r}_1 \times \mathbf{r}_2'$ — cross product rule (ORDER MATTERS)

Did You Know?

Newton's second law of motion $\mathbf{F} = m\mathbf{a}$ is a vector equation involving the second derivative of a position vector: $\mathbf{F} = m \cdot \mathbf{r}''(t)$. This is the starting point for orbital mechanics, robotics, and vehicle dynamics. ISRO's PSLV (Polar Satellite Launch Vehicle) trajectory computation uses exactly this vector calculus framework — computing the position vector $\mathbf{r}(t)$ of the spacecraft at every instant using numerical integration of $\mathbf{F} = m\mathbf{a}$, where $\mathbf{F}$ includes thrust, gravity, and atmospheric drag as vector quantities changing with time and position.

4.9 Scalar and Vector Point Functions

In vector calculus, we study functions defined at every point in a region of space. Such functions are called point functions (or fields). There are two fundamental types, depending on whether the function assigns a scalar or a vector to each point.

Scalar Point Function (Scalar Field)

A scalar point function assigns a single real number (a scalar) to each point in a region of space. Formally, if to every point $P(x, y, z)$ in some region $D \subseteq \mathbb{R}^3$ there corresponds a unique scalar $\varphi(x, y, z)$, then φ is a scalar point function (or scalar field) on D .

Examples of scalar fields in physics and engineering:

- Temperature distribution: $T(x, y, z)$ — the temperature at each point in a room.
- Pressure field: $P(x, y, z)$ — pressure at each point in a fluid.
- Electric potential: $V(x, y, z)$ — potential at each point in an electric field.
- Density: $\rho(x, y, z)$ — mass density at each point in a solid body.

Vector Point Function (Vector Field)

A vector point function assigns a vector to each point in a region. If to every point $P(x, y, z)$ there corresponds a vector $F(x, y, z) = F_1(x, y, z)\hat{i} + F_2(x, y, z)\hat{j} + F_3(x, y, z)\hat{k}$, then F is a vector point function (or vector field).

Examples of vector fields in physics and engineering:

- Gravitational field: $g(x, y, z)$ — gravitational force per unit mass at each point.
- Electric field: $E(x, y, z)$ — force per unit positive charge at each point.
- Magnetic field: $B(x, y, z)$ — magnetic flux density at each point.
- Velocity field of a fluid: $v(x, y, z)$ — velocity of fluid at each point in flow.
- Wind velocity field: $W(x, y, z)$ — wind speed and direction at each location.

The Del Operator ∇

The del (or nabla) operator ∇ is the fundamental differential operator of vector calculus.

In three-dimensional Cartesian coordinates, it is defined as:

The del (nabla) operator: $\nabla = (\partial/\partial x)\hat{i} + (\partial/\partial y)\hat{j} + (\partial/\partial z)\hat{k}$

The del operator is not a function or a vector by itself — it is a symbolic operator that acts on scalar or vector fields to produce new fields. The three main operations are the gradient ($\nabla\phi$), divergence ($\nabla\cdot F$), and curl ($\nabla\times F$), each studied in the following sub-units.

4.10 Gradient of a Scalar Field

The gradient is the most natural differential operation on a scalar field. It takes a scalar field $\phi(x, y, z)$ — a function that assigns a number to each point — and produces a vector field $\nabla\phi$ that points in the direction of greatest increase of ϕ , with magnitude equal to the rate of increase in that direction.

Intuitively: if ϕ represents temperature, then $\nabla\phi$ at any point points toward the hottest nearby region. If ϕ is altitude on a terrain map, $\nabla\phi$ points straight uphill at each location.

Definition and Formula

For a scalar field $\phi(x, y, z)$, the gradient is:

Gradient: $\nabla\phi = (\partial\phi/\partial x)\hat{i} + (\partial\phi/\partial y)\hat{j} + (\partial\phi/\partial z)\hat{k}$

Properties of the Gradient

- Direction: $\nabla\phi$ points in the direction of maximum rate of increase of ϕ .
- Magnitude: $|\nabla\phi|$ equals the maximum rate of change of ϕ at that point.
- Perpendicularity: $\nabla\phi$ is always perpendicular to the level surfaces $\phi = \text{constant}$.

- Directional derivative: The rate of change of φ in direction $\hat{a}$ is $(\nabla\varphi) \cdot \hat{a} = |\nabla\varphi|\cos\theta$.

Directional Derivative

The directional derivative of φ in the direction of a unit vector $\hat{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is:

Directional derivative: $D_{\hat{a}}\varphi = \nabla\varphi \cdot \hat{a} = (\partial\varphi/\partial x)a_1 + (\partial\varphi/\partial y)a_2 + (\partial\varphi/\partial z)a_3$

This measures how fast φ changes as you move in direction $\hat{a}$. The maximum directional derivative equals $|\nabla\varphi|$, achieved in the direction $\hat{a} = \nabla\varphi/|\nabla\varphi|$.

Worked Example: Computing the Gradient

If $\varphi = x^2y + yz^3$, find $\nabla\varphi$ and the directional derivative at $(1, 2, 1)$ in direction of $\hat{i} + \hat{j} - \hat{k}$.

Step 1: Compute partial derivatives:

$$\partial\varphi/\partial x = 2xy, \quad \partial\varphi/\partial y = x^2 + z^3, \quad \partial\varphi/\partial z = 3yz^2$$

Step 2: $\nabla\varphi = 2xy\hat{i} + (x^2 + z^3)\hat{j} + 3yz^2\hat{k}$

Step 3: At $(1, 2, 1)$: $\nabla\varphi = 2(1)(2)\hat{i} + (1 + 1)\hat{j} + 3(2)(1)\hat{k} = 4\hat{i} + 2\hat{j} + 6\hat{k}$

Step 4: Direction vector: $d = \hat{i} + \hat{j} - \hat{k}$. Unit vector: $\hat{a} = (\hat{i} + \hat{j} - \hat{k})/\sqrt{3}$

Step 5: Directional derivative = $\nabla\varphi \cdot \hat{a} = (4 \cdot 1 + 2 \cdot 1 + 6 \cdot (-1))/\sqrt{3} = (4+2-6)/\sqrt{3} = 0/\sqrt{3} = 0$

The directional derivative is 0, meaning φ has no change in the direction $\hat{i} + \hat{j} - \hat{k}$ at $(1, 2, 1)$.

$|\nabla\varphi| = \sqrt{(16 + 4 + 36)} = \sqrt{56} = 2\sqrt{14}$ (maximum rate of change).

4.11 Divergence of a Vector Field

The divergence of a vector field measures the tendency of the field to "spread out" or "converge" at each point — it is a scalar measure of how much the vector field is acting as a source or a sink at that location. If you think of the vector field as describing fluid velocity, the divergence at a point tells you the net rate at which fluid is flowing away from (positive) or toward (negative) that point per unit volume.

Definition and Formula

For a vector field $F = F_1(x,y,z)\hat{i} + F_2(x,y,z)\hat{j} + F_3(x,y,z)\hat{k}$, the divergence is:

Divergence: $\nabla \cdot F = \partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z$

The notation $\nabla \cdot F$ is read "del dot F" or "div F". Note that divergence takes a vector field as input and produces a scalar field as output.

Physical Interpretation

- $\nabla \cdot F > 0$ at a point: the field has a source at that point — flux is flowing outward.
- $\nabla \cdot F < 0$ at a point: the field has a sink at that point — flux is flowing inward.
- $\nabla \cdot F = 0$ everywhere: the field is solenoidal (source-free) — no net creation or destruction of flux. This applies to magnetic fields (Gauss's law for magnetism: $\nabla \cdot B = 0$) and incompressible fluid flow.

Worked Example: Computing Divergence

Find $\nabla \cdot F$ for: (i) $F = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ (ii) $F = xyz\hat{i} + (x^2+y^2)\hat{j} + (y^2z)\hat{k}$

Step 1: (i) $F_1 = x^2, F_2 = y^2, F_3 = z^2$

$$\nabla \cdot F = \partial(x^2)/\partial x + \partial(y^2)/\partial y + \partial(z^2)/\partial z = 2x + 2y + 2z$$

At origin (0,0,0): $\nabla \cdot F = 0$. At point (1,1,1): $\nabla \cdot F = 6$ (positive — source).

Step 2: (ii) $F_1 = xyz, F_2 = x^2+y^2, F_3 = y^2z$

$$\partial F_1 / \partial x = yz, \quad \partial F_2 / \partial y = 2y, \quad \partial F_3 / \partial z = y^2$$

$$\nabla \cdot \mathbf{F} = yz + 2y + y^2 = y(z + 2 + y)$$

4.12 Curl of a Vector Field

The curl of a vector field measures the rotational tendency of the field at each point. While divergence measures source/sink behaviour (radial expansion or contraction), curl measures the tendency of the field to circulate around a point — like a tiny paddle wheel placed in a fluid: the curl tells you how fast and in what direction the paddle wheel would spin.

The curl of a vector field $\mathbf{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ is a vector field given by:

$$\text{Curl: } \nabla \times \mathbf{F} = (\partial F_3 / \partial y - \partial F_2 / \partial z) \cdot \hat{i} - (\partial F_3 / \partial x - \partial F_1 / \partial z) \cdot \hat{j} + (\partial F_2 / \partial x - \partial F_1 / \partial y) \cdot \hat{k}$$

A convenient mnemonic is to write it as the formal 3×3 determinant:

$$\begin{aligned} & \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix} \\ \nabla \times \mathbf{F} = & \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix} \end{aligned}$$

Expanding this determinant along the first row gives exactly the component formula above. The notation $\nabla \times \mathbf{F}$ is read "del cross $\mathbf{F}$ " or "curl $\mathbf{F}$ ". Curl takes a vector field and returns a vector field.

Physical Interpretation

- $\nabla \times \mathbf{F} \neq 0$ at a point: the field has rotational behaviour — circulation around that point.
- $\nabla \times \mathbf{F} = 0$ everywhere: the field is irrotational — no rotation at any point.
- For a velocity field $\mathbf{v}$: $\nabla \times \mathbf{v} = 2\boldsymbol{\omega}$ where $\boldsymbol{\omega}$ is the local angular velocity of fluid rotation.

- In electrostatics: $\nabla \times \mathbf{E} = 0$ (electrostatic fields are irrotational — they are conservative).
- In electrodynamics (Faraday): $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$.

Worked Example: Computing Curl

Find $\nabla \times \mathbf{F}$ for: (i) $\mathbf{F} = (2xy)\hat{i} + (x^2 + z^2)\hat{j} + (2yz)\hat{k}$ (ii) $\mathbf{F} = y^2\hat{i} + 2xy\hat{j} + z\hat{k}$

Step 1: (i) $F_1 = 2xy$, $F_2 = x^2 + z^2$, $F_3 = 2yz$

$\hat{i}$ -component: $\partial F_3 / \partial y - \partial F_2 / \partial z = 2z - 2z = 0$

$\hat{j}$ -component: $-(\partial F_3 / \partial x - \partial F_1 / \partial z) = -(0 - 0) = 0$

$\hat{k}$ -component: $\partial F_2 / \partial x - \partial F_1 / \partial y = 2x - 2x = 0$

$\nabla \times \mathbf{F} = 0$. This field is irrotational!

Step 2: (ii) $F_1 = y^2$, $F_2 = 2xy$, $F_3 = z$

$\hat{i}$ -component: $\partial(z) / \partial y - \partial(2xy) / \partial z = 0 - 0 = 0$

$\hat{j}$ -component: $-(\partial(z) / \partial x - \partial(y^2) / \partial z) = -(0 - 0) = 0$

$\hat{k}$ -component: $\partial(2xy) / \partial x - \partial(y^2) / \partial y = 2y - 2y = 0$

$\nabla \times \mathbf{F} = 0$. This field is also irrotational!

Analysing Misconceptions: Divergence and Curl Measure the Same Thing

✗ MYTH: Both divergence and curl measure how much a vector field is "spreading out", so they give the same information.

✓ REALITY: Divergence and curl measure fundamentally different types of behaviour. Divergence ($\nabla \cdot \mathbf{F}$) is a scalar that measures the net outflow of the field per unit volume — source and sink behaviour (radial expansion/compression). Curl ($\nabla \times \mathbf{F}$) is a vector that measures the rotational tendency around each point — circulation. A field can have nonzero divergence but zero curl (like gravity near a point mass), or zero divergence but nonzero curl (like the magnetic field around a

current-carrying wire), or both nonzero or both zero. They are independent quantities.

4.13 The Laplacian Operator

The Laplacian is one of the most important differential operators in all of mathematics and physics. It is defined as the divergence of the gradient of a scalar field — applying $\nabla \cdot$ to $\nabla \phi$. This composite operation produces a scalar field from a scalar field:

Laplacian: $\nabla^2 \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$

The notation $\nabla^2 \phi$ (read "del-squared phi" or "Laplacian of phi") is universally standard. The Laplacian can also be applied to each component of a vector field F , producing the vector Laplacian $\nabla^2 F$.

Physical Significance of the Laplacian

The Laplacian appears in the fundamental partial differential equations governing physical phenomena:

- Laplace's equation: $\nabla^2 \phi = 0$. Solutions (called harmonic functions) describe electrostatic potential in charge-free regions, steady-state temperature distributions, and ideal fluid flow.
- Poisson's equation: $\nabla^2 \phi = \rho/\epsilon_0$. Governs electrostatic potential in regions containing charge.
- Heat equation: $\partial T/\partial t = \alpha \nabla^2 T$. Governs heat diffusion through a solid body.
- Wave equation: $\partial^2 u/\partial t^2 = c^2 \nabla^2 u$. Governs propagation of sound, light, and seismic waves.
- Schrödinger equation: $-(\hbar^2/2m) \nabla^2 \psi + V\psi = E\psi$. Governs quantum mechanical behaviour.

Harmonic Functions

A function φ satisfying Laplace's equation $\nabla^2\varphi = 0$ is called a harmonic function. The study of harmonic functions is a rich area of mathematics with deep connections to complex analysis (via the real and imaginary parts of analytic functions) and to potential theory.

Worked Example: Computing the Laplacian

Compute $\nabla^2\varphi$ for: (i) $\varphi = x^2 + y^2 + z^2$ (ii) $\varphi = e^{(x+y)} \sin z$ (iii) $\varphi = \ln\sqrt{x^2+y^2}$

Step 1: (i) $\varphi = x^2 + y^2 + z^2$

$$\partial^2\varphi/\partial x^2 = 2, \quad \partial^2\varphi/\partial y^2 = 2, \quad \partial^2\varphi/\partial z^2 = 2$$

$$\nabla^2\varphi = 2 + 2 + 2 = 6$$

Step 2: (ii) $\varphi = e^{(x+y)} \sin z$

$$\partial\varphi/\partial x = e^{(x+y)} \sin z, \quad \partial^2\varphi/\partial x^2 = e^{(x+y)} \sin z$$

$$\partial^2\varphi/\partial y^2 = e^{(x+y)} \sin z$$

$$\partial^2\varphi/\partial z^2 = -e^{(x+y)} \sin z$$

$$\nabla^2\varphi = e^{(x+y)} \sin z + e^{(x+y)} \sin z - e^{(x+y)} \sin z = e^{(x+y)} \sin z$$

Step 3: (iii) $\varphi = \ln\sqrt{x^2+y^2} = (1/2)\ln(x^2+y^2)$

$$\partial\varphi/\partial x = x/(x^2+y^2), \quad \partial^2\varphi/\partial x^2 = (y^2-x^2)/(x^2+y^2)^2$$

$$\partial^2\varphi/\partial y^2 = (x^2-y^2)/(x^2+y^2)^2$$

$$\nabla^2\varphi = (y^2-x^2)/(x^2+y^2)^2 + (x^2-y^2)/(x^2+y^2)^2 = 0$$

$\varphi = \ln\sqrt{x^2+y^2}$ is a harmonic function (satisfies $\nabla^2\varphi = 0$ for $x, y \neq 0$).

4.14 Solenoidal and Irrotational Vector Fields

The two most important special classes of vector fields in physics and engineering are defined by whether the divergence or curl is zero everywhere. These conditions have deep physical interpretations and enable powerful mathematical simplifications.

Solenoidal Vector Fields (Divergence-Free)

A vector field F is called solenoidal (or incompressible or divergence-free) if its divergence is zero everywhere in the region of interest:

Solenoidal condition: $\nabla \cdot F = 0$ everywhere $\Leftrightarrow F$ is solenoidal

Physical meaning: a solenoidal field has no sources or sinks — the field lines neither start nor end (they form closed loops, or enter/exit the boundary). By the Divergence Theorem, this means the net flux through any closed surface in the region is zero.

Examples of solenoidal fields:

- Magnetic field B : Maxwell's equation $\nabla \cdot B = 0$ states that magnetic monopoles do not exist. Magnetic field lines always form closed loops.
- Incompressible fluid velocity field v : $\nabla \cdot v = 0$ means fluid is neither created nor destroyed (e.g., water, blood flow in arteries).
- Induction magnetic field: the field induced by a current-carrying coil satisfies $\nabla \cdot B = 0$.

Irrotational Vector Fields (Curl-Free)

A vector field F is called irrotational (or conservative or curl-free) if its curl is zero everywhere:

Irrotational condition: $\nabla \times F = 0$ everywhere $\Leftrightarrow F$ is irrotational

Physical meaning: an irrotational field has no rotational circulation — a tiny paddle wheel placed in the field would not rotate. Equivalently, an irrotational field is a conservative field — it can be expressed as the gradient of a scalar potential: $F = \nabla\phi$.

Examples of irrotational fields:

- Gravitational field $g = -GM/r^2 \cdot \hat{r}$: $\nabla \times g = 0$ (gravity is conservative — work is path-independent).
- Electrostatic field E : $\nabla \times E = 0$ in static conditions — this is why electric potential V exists such that $E = -\nabla V$.
- Velocity field of irrotational fluid flow (ideal fluids): $\nabla \times v = 0$ enables use of velocity potential.

Connection Between the Two: The Scalar Potential

If F is irrotational ($\nabla \times F = 0$), then there exists a scalar function ϕ (called the scalar potential) such that $F = \nabla\phi$. Conversely, if F is solenoidal ($\nabla \cdot F = 0$), there exists a vector function A (the vector potential) such that $F = \nabla \times A$. These results are versions of the Helmholtz Decomposition Theorem.

Worked Example: Testing Solenoidal and Irrotational Properties

For $F = (y^2+z)\hat{i} + (2xy)\hat{j} + (x)\hat{k}$, determine whether F is solenoidal and/or irrotational.

Step 1: Test solenoidal (compute $\nabla \cdot F$):

$$F_1 = y^2+z, \quad F_2 = 2xy, \quad F_3 = x$$

$$\partial F_1 / \partial x = 0, \quad \partial F_2 / \partial y = 2x, \quad \partial F_3 / \partial z = 0$$

$$\nabla \cdot F = 0 + 2x + 0 = 2x \neq 0 \text{ (unless } x = 0\text{)}$$

F is NOT solenoidal (has nonzero divergence for $x \neq 0$).

Step 2: Test irrotational (compute $\nabla \times F$):

$$\hat{i}\text{-component: } \partial(x)/\partial y - \partial(2xy)/\partial z = 0 - 0 = 0$$

$$\hat{j}\text{-component: } -[\partial(x)/\partial x - \partial(y^2+z)/\partial z] = -[1 - 1] = 0$$

$$\hat{k}\text{-component: } \partial(2xy)/\partial x - \partial(y^2+z)/\partial y = 2y - 2y = 0$$

$$\nabla \times \mathbf{F} = 0. \mathbf{F} \text{ IS irrotational! } \checkmark$$

Since $\mathbf{F}$ is irrotational, there exists a scalar potential φ such that $\mathbf{F} = \nabla \varphi$.

Worked Example: Finding the Scalar Potential

Find the scalar potential φ for the irrotational field $\mathbf{F} = (2xyz)\hat{i} + (x^2z)\hat{j} + (x^2y)\hat{k}$.

Step 1: Verify irrotational: $\nabla \times \mathbf{F} = 0$ (verify each component).

$$\hat{i}: \partial(x^2y)/\partial y - \partial(x^2z)/\partial z = x^2 - x^2 = 0 \quad \checkmark$$

$$\hat{j}: -[\partial(x^2y)/\partial x - \partial(2xyz)/\partial z] = -[2xy - 2xy] = 0 \quad \checkmark$$

$$\hat{k}: \partial(x^2z)/\partial x - \partial(2xyz)/\partial y = 2xz - 2xz = 0 \quad \checkmark$$

Step 2: Find φ from $\partial\varphi/\partial x = F_1 = 2xyz$:

$$\varphi = \int 2xyz \, dx = x^2yz + g(y,z) \quad \text{where } g \text{ depends only on } y,z$$

Step 3: Use $\partial\varphi/\partial y = F_2 = x^2z$:

$$\partial\varphi/\partial y = x^2z + \partial g/\partial y = x^2z \implies \partial g/\partial y = 0 \implies g = h(z)$$

Step 4: Use $\partial\varphi/\partial z = F_3 = x^2y$:

$$\partial\varphi/\partial z = x^2y + h'(z) = x^2y \implies h'(z) = 0 \implies h = \text{constant}$$

Step 5: Scalar potential: $\varphi = x^2yz + C$ (verify: $\nabla(x^2yz) = 2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k} = \mathbf{F}$ $\checkmark$)

 Let's Check

Q.No	Type	Question
1	MCQ	The gradient of a scalar field ϕ points in the direction of:
2	MCQ	A vector field F is solenoidal if:
3	Fill in Blank	The Laplacian $\nabla^2\phi$ equals _____ + $\partial^2\phi/\partial y^2$ + $\partial^2\phi/\partial z^2$.
4	Fill in Blank	An irrotational vector field satisfies $\nabla \times F =$ _____.
5	True/False	The magnetic field B is solenoidal because $\nabla \cdot B = 0$ (no magnetic monopoles exist).
6	True/False	If $\nabla \times F = 0$ everywhere, then F must be the gradient of some scalar field ϕ .

Answers:

Q.No	Answer
1	c) Maximum increase of ϕ — the gradient vector always points uphill (toward higher values of ϕ).
2	b) $\nabla \cdot F = 0$ — zero divergence means no sources or sinks (incompressible field).
3	$\partial^2\phi/\partial x^2$
4	0 (the zero vector)
5	True — Maxwell's equation $\nabla \cdot B = 0$ holds universally; magnetic field lines form closed loops.
6	True — if $\nabla \times F = 0$ (irrotational/conservative), a scalar potential ϕ exists such that $F = \nabla\phi$.

■ Case Study: Vector Calculus in Indian Power Grid — Electromagnetic Field Analysis at POWERGRID

Background:

POWERGRID Corporation of India (Power Grid Corporation of India Limited) manages one of the world's largest electricity transmission networks — over 1,72,000 circuit kilometres of high-voltage transmission lines, connecting generation centres to distribution hubs across India. The engineering analysis of this network relies directly on the vector calculus concepts studied in this module: gradient, divergence, curl, and Laplacian.

Electromagnetic Fields in Transmission Lines:

A high-voltage transmission line carrying alternating current generates both electric and magnetic fields. The complete description of these fields is governed by Maxwell's four equations of electromagnetism, each expressed in terms of the vector operators studied in this module:

- Gauss's Law for electricity: $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$ (divergence of electric field = charge density / permittivity). POWERGRID engineers use this to compute the electric field distribution around their 765 kV ultra-high voltage lines and ensure it stays within safe limits for nearby residents (< 5 kV/m at ground level per PGCIL safety standards).
- Gauss's Law for magnetism: $\nabla \cdot \mathbf{B} = 0$ (magnetic field is solenoidal — no magnetic monopoles). This guarantees that every magnetic field line forms a closed loop, critical for transformer core design at POWERGRID's Bina 765/400 kV substation in Madhya Pradesh.
- Faraday's Law: $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ (curl of electric field = rate of change of magnetic flux). This is the mathematical basis of electromagnetic induction

— how every transformer in India's grid works. The curl operation quantifies the rotational tendency of E that drives induced currents.

- Ampère-Maxwell Law: $\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \partial E / \partial t$ (curl of magnetic field = current density + displacement current). Used by POWERGRID's protection engineers to compute the magnetic field that drives protective relay operation during fault conditions.

Laplacian in Power System Analysis:

In steady-state (no time variation), the electric potential $V(x, y, z)$ in charge-free regions satisfies Laplace's equation: $\nabla^2 V = 0$. POWERGRID's research centre in Manesar (Haryana) uses numerical solutions of this equation — solved by finite element methods — to design insulator profiles for 765 kV towers, optimising the shape to achieve uniform potential distribution and prevent corona discharge.

Integration in Power Calculations:

The total energy stored in the electric field of a capacitor bank (used for reactive power compensation at POWERGRID substations) is computed as the volume integral $\iiint (1/2) \epsilon_0 |E|^2 dV$. This triple integral — a direct extension of the definite integration studied in Sub-unit 4.6 — requires computing $\int E^2 dV$ over the capacitor's field region. Similarly, the work done in moving charge through the electric field uses line integrals, which are computed by reducing to the definite integrals of this module.

(Source: POWERGRID Corporation Annual Technical Report; PGCIL EMF safety guidelines based on ICNIRP standards.)

Module Summary

Module 4 has developed two powerful and deeply connected branches of advanced calculus: integral calculus and vector calculus. Together, they provide the mathematical machinery for computing accumulated quantities and analysing the behaviour of physical fields in three-dimensional space.

In integral calculus, the journey began with the recognition that integration is the inverse of differentiation — with the constant of integration reflecting the non-uniqueness of the antiderivative. A systematic library of standard results was established and then extended through four fundamental techniques:

- Substitution (u-substitution): Reverse of the chain rule. Identify a sub-expression whose derivative also appears in the integrand, substitute, simplify, integrate, and back-substitute.
- Trigonometric identities: Rewrite powers and products of trig functions using half-angle, double-angle, and Pythagorean identities before integrating.
- Partial fractions: Decompose a proper rational function into simpler fractions (linear and quadratic denominators) and integrate each term using logarithms and arctangents.
- Integration by parts: $\int u \, dv = uv - \int v \, du$. Use ILATE to choose u ; suitable for products of different function families.

The Fundamental Theorem of Calculus connects indefinite and definite integrals: $\int_a^b f(x)dx = F(b) - F(a)$. Properties of definite integrals — linearity, additivity, symmetry (even/odd functions), and the king's property — simplify evaluation. Reduction formulae provide efficient recursive formulas for $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$, with the Wallis formula giving elegant closed-form results over $[0, \pi/2]$.

In vector calculus, vector functions of a scalar parameter describe particle trajectories, with derivatives giving velocity and second derivatives giving acceleration — connecting directly to Newton's law $F = ma$. Scalar and vector fields assign numbers

and vectors to every point in space, and the del operator ∇ acts on them to produce new fields.

The three fundamental operators:

- Gradient $\nabla\phi$: acts on a scalar field; produces a vector field pointing in the direction of maximum increase. Magnitude = maximum rate of change. Perpendicular to level surfaces.
- Divergence $\nabla\cdot\mathbf{F}$: acts on a vector field; produces a scalar measuring the net outflow per unit volume. Zero divergence $\Leftrightarrow$ solenoidal (no sources or sinks).
- Curl $\nabla\times\mathbf{F}$: acts on a vector field; produces a vector measuring rotational tendency. Zero curl $\Leftrightarrow$ irrotational (conservative — has a scalar potential).
- Laplacian $\nabla^2\phi = \nabla\cdot(\nabla\phi) = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$: combines gradient and divergence; zero Laplacian defines harmonic functions.

Glossary

Term	Definition
Indefinite Integral	$\int f(x)dx = F(x) + C$, where $F'(x) = f(x)$. The family of all antiderivatives of f , differing only by an additive constant.
Integration by Substitution	Technique replacing the variable with $u = g(x)$ to simplify an integral. Reverses the chain rule.
Integration by Parts	Formula $\int u dv = uv - \int v du$, reversing the product rule. ILATE guides the choice of u .
Partial Fractions	Method of decomposing a proper rational function into a sum of simpler fractions for easier integration.
Definite Integral	$\int [a \text{ to } b] f(x)dx = F(b) - F(a)$. Computes the net signed area under f between $x=a$ and $x=b$.
Reduction Formula	Recursive formula expressing $\int \sin^n x dx$ or $\int \cos^n x dx$ in terms of $\int \sin^{n-2} x dx$ or $\int \cos^{n-2} x dx$.
Vector Field	A function that assigns a vector to each point in a region of space. E.g., $F(x,y,z) = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$.
Gradient ($\nabla\phi$)	The vector field of partial derivatives of a scalar ϕ : $\nabla\phi = (\partial\phi/\partial x)\hat{i} + (\partial\phi/\partial y)\hat{j} + (\partial\phi/\partial z)\hat{k}$. Points in direction of maximum increase.
Divergence ($\nabla \cdot F$)	Scalar measure of net outflow of vector field F : $\nabla \cdot F = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$. Zero divergence $\Rightarrow$ solenoidal.
Curl ($\nabla \times F$)	Vector measure of rotational tendency of field F . Zero curl $\Rightarrow$ irrotational (conservative) field.

Self Assessment Questions

Section A: Multiple Choice Questions (MCQ)

Choose the most appropriate answer for each question.

1. $\int x^3 dx$ equals:

- a) $3x^2$
- b) $x^4/4 + C$
- c) $x^4 + C$
- d) $4x^4 + C$

2. The substitution to evaluate $\int \sin x \cdot \cos x dx$ is:

- a) $u = \sin x, du = \cos x dx$
- b) $u = \cos x, du = \sin x dx$
- c) Either a) or b) works
- d) No substitution needed

3. $\int (1/x) dx$ equals:

- a) $1/x^2 + C$
- b) $\ln|x| + C$
- c) $x^{-1} + C$
- d) $-1/x^2 + C$

4. In the ILATE rule for integration by parts, which function type is chosen first as u ?

- a) Exponential
- b) Trigonometric
- c) Logarithmic
- d) Algebraic

5. $\int [0 \text{ to } \pi] \sin x dx$ equals:

- a) 0
- b) 1

-
- c) 2
- d) π
6. The Wallis formula gives $\int [0 \text{ to } \pi/2] \sin^4 x \, dx$ as:
- a) $\pi/4$
- b) $3\pi/16$
- c) $\pi/8$
- d) $3\pi/8$
7. For partial fractions of $(x+1)/[(x-1)(x-2)]$, the form is:
- a) $A/(x-1)(x-2)$
- b) $A/(x-1) + B/(x-2)$
- c) $(Ax+B)/(x-1)(x-2)$
- d) $A + B/(x-1)$
8. A vector function $\mathbf{r}(t)$ has derivative $\mathbf{r}'(t)$ representing:
- a) Speed
- b) Acceleration
- c) Velocity vector
- d) Position
9. The gradient $\nabla\phi$ of a scalar field ϕ is:
- a) A scalar
- b) A vector pointing in direction of maximum increase
- c) Always parallel to the x-axis
- d) Equal to $\nabla^2\phi$
10. The divergence $\nabla \cdot \mathbf{F}$ of a vector field $\mathbf{F} = x\hat{i} + y\hat{j} + z\hat{k}$ is:
- a) $\hat{i} + \hat{j} + \hat{k}$
- b) $x+y+z$
- c) 3

d) 0

11. A solenoidal vector field satisfies:

a) $\nabla \times \mathbf{F} = 0$

b) $\nabla \cdot \mathbf{F} = 0$

c) $\nabla^2 \mathbf{F} = 0$

d) $\nabla \mathbf{F} = 0$

12. An irrotational (conservative) vector field satisfies:

a) $\nabla \cdot \mathbf{F} = 0$

b) $\nabla^2 \mathbf{F} = 0$

c) $\nabla \times \mathbf{F} = 0$

d) $\mathbf{F} = \nabla^2 \phi$

13. The Laplacian $\nabla^2 \phi$ of a scalar field is:

a) $\nabla \times (\nabla \phi)$

b) $\nabla(\nabla \cdot \phi)$

c) $\partial^2 \phi / \partial x^2 + \partial^2 \phi / \partial y^2 + \partial^2 \phi / \partial z^2$

d) $|\nabla \phi|^2$

14. A function satisfying $\nabla^2 \phi = 0$ is called:

a) Solenoidal

b) Irrotational

c) Harmonic

d) Parametric

15. Which of the following physical fields satisfies $\nabla \cdot \mathbf{B} = 0$?

a) Electric field $\mathbf{E}$ near a charge

b) Gravitational potential V

c) Magnetic field $\mathbf{B}$ always

d) Temperature gradient ∇T

16. For $\int x \ln x \, dx$, the correct choice of u and dv by ILATE is:

- a) $u = x, dv = \ln x \, dx$
- b) $u = \ln x, dv = x \, dx$
- c) $u = 1, dv = x \ln x \, dx$
- d) $u = x \ln x, dv = dx$

17. Using the identity $\sin^2 x = (1 - \cos 2x)/2$, $\int \sin^2 x \, dx$ equals:

- a) $-\sin^2 x/2 + C$
- b) $x/2 - \sin(2x)/4 + C$
- c) $\cos(2x)/2 + C$
- d) $\sin(2x)/4 + C$

18. For the reduction formula $I_n = \int [0 \text{ to } \pi/2] \sin^n x \, dx$, the value of I_1 is:

- a) 0
- b) $\pi/2$
- c) 1
- d) π

19. The scalar potential ϕ of an irrotational field F satisfies:

- a) $\nabla^2 \phi = F$
- b) $\nabla \cdot \phi = F$
- c) $\nabla \phi = F$
- d) $\nabla \times \phi = F$

20. The directional derivative of ϕ in direction $\hat{a}$ equals:

- a) $\nabla \phi \times \hat{a}$
- b) $\nabla \phi \cdot \hat{a}$
- c) $|\nabla \phi|$
- d) $\nabla^2 \phi \cdot \hat{a}$

Section B: True / False

State whether each statement is True or False.

1. $\int [a \text{ to } b] f(x)dx = -\int [b \text{ to } a] f(x)dx$ for any function f .
2. The curl of a gradient is always the zero vector: $\nabla \times (\nabla \phi) = 0$.
3. The divergence of a curl is always non-zero: $\nabla \cdot (\nabla \times F) \neq 0$.
4. For an even function $f(x)$, $\int [-a \text{ to } a] f(x)dx = 0$.
5. Every irrotational vector field is the gradient of some scalar potential ϕ (in a simply connected region).

Section C: Short Answer / Problem Solving

Answer each question with full working, clearly showing all steps.

1. Evaluate the following integrals with full working: (i) $\int x^2 \cdot e^{-x} dx$ using integration by parts, (ii) $\int (2x+3)/[(x^2+1)(x+1)] dx$ using partial fractions, (iii) $\int [0 \text{ to } \pi/2] \sin^3 x \cos^2 x dx$.
2. Derive the reduction formula for $I_n = \int [0 \text{ to } \pi/2] \cos^n x dx$ using integration by parts. Use it to evaluate I_4 and I_5 .
3. A particle moves along a curve with position vector $r(t) = (t^2 - 1)\hat{i} + (2t)\hat{j} + (t^3/3)\hat{k}$. Find: (i) velocity vector $v(t)$, (ii) speed $|v|$ at $t = 1$, (iii) acceleration vector $a(t)$, (iv) unit tangent vector $\hat{T}$ at $t = 1$.
4. For the scalar field $\phi = x^2yz + xz^2$, find: (i) $\nabla \phi$ at the point $(1, 1, 2)$, (ii) the directional derivative in the direction $2\hat{i} - \hat{j} + 2\hat{k}$ at $(1,1,2)$, (iii) the maximum rate of increase and the direction in which it occurs.
5. For the vector field $F = (x^2y)\hat{i} + (y^2z)\hat{j} + (z^2x)\hat{k}$: (i) compute $\nabla \cdot F$, (ii) compute $\nabla \times F$, (iii) determine whether F is solenoidal or irrotational at the point $(1, 1, 1)$.

Self Assessment — Answer Keys

Section A — MCQ

Q.No	Answer	Option Text
1	B	b) $x^4/4 + C$
2	C	c) Either a) or b) works
3	B	b) $\ln x + C$
4	C	c) Logarithmic
5	C	c) 2
6	B	b) $3\pi/16$
7	B	b) $A/(x-1) + B/(x-2)$
8	C	c) Velocity vector
9	B	b) A vector pointing in direction of maximum increase
10	C	c) 3
11	B	b) $\nabla \cdot \mathbf{F} = 0$
12	C	c) $\nabla \times \mathbf{F} = 0$
13	C	c) $\partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$
14	C	c) Harmonic
15	C	c) Magnetic field $\mathbf{B}$ always
16	B	b) $u = \ln x$, $dv = x dx$
17	B	b) $x/2 - \sin(2x)/4 + C$
18	C	c) 1

19	C	c) $\nabla\varphi = F$
20	B	b) $\nabla\varphi \cdot \hat{a}$

Section B — True / False Answer Key

Q.No	Answer	Notes
1	True	Correct.
2	True	Correct.
3	False	False — $\nabla \cdot (\nabla \times F) = 0$ always (the divergence of any curl is identically zero — a fundamental vector identity).
4	False	False — for an even function the integral over a symmetric interval equals $2 \int [0 \text{ to } a] f(x) dx$ (not zero). Zero integral applies to ODD functions.
5	True	Correct.

Section C — Short Answer Model Solutions

Q.No	Model Solution
1	(i) $\int x^2 e^{-x} dx$. ILATE: $u = x^2$, $dv = e^{-x} dx$. $v = -e^{-x}$. $= -x^2 e^{-x} - \int -e^{-x} \cdot 2x dx = -x^2 e^{-x} + 2 \int x e^{-x} dx$. Again: $u = x, dv = e^{-x} dx \rightarrow = -x^2 e^{-x} + 2[-x e^{-x} + \int e^{-x} dx] = -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + C = -e^{-x}(x^2 + 2x + 2) + C$. (ii) Decompose: $A/(x+1) + (Bx+C)/(x^2+1)$. $2x+3 = A(x^2+1) + (Bx+C)(x+1)$. $x = -1$: $1 = 2A \rightarrow A = 1/2$. $x = 0$: $3 = A + C \rightarrow C = 5/2$. $x = 1$: $5 = 2A + 2B + 2C \rightarrow B = 1/2$. $\int = (1/2)\ln x+1 + (1/4)\ln(x^2+1) + (5/2)\tan^{-1}x + C$. (iii) $\int [0 \text{ to } \pi/2] \sin^3 x \cos^2 x dx$.

	$u = \cos x: \int \sin^2 x \cdot \cos^2 x \cdot \sin x \, dx = \int (1-u^2)u^2(-du) \text{ from } 1 \text{ to } 0 = \int [0 \text{ to } 1](u^2-u^4)du = [u^3/3 - u^5/5]_0^1 = 1/3 - 1/5 = 2/15.$
2	Deriving I_n: $u = \cos^{n-1} x$, $dv = \cos x \, dx \rightarrow v = \sin x$, $du = (n-1)\cos^{n-2} x(-\sin x)dx$. $I_n = [\cos^{n-1} x \cdot \sin x]_0^{\pi/2} + (n-1) \int [0 \text{ to } \pi/2] \cos^{n-2} x \cdot \sin^2 x \, dx = 0 + (n-1) \int \cos^{n-2} x (1 - \cos^2 x) dx = (n-1)I_{n-2} - (n-1)I_n$. So $nI_n = (n-1)I_{n-2} \rightarrow I_n = [(n-1)/n]I_{n-2}$. $I_4 = (3/4) \cdot (1/2) \cdot (\pi/2) = 3\pi/16$. $I_5 = (4/5) \cdot (2/3) \cdot 1 = 8/15$.
3	(i) $v(t) = r'(t) = 2t\hat{i} + 2\hat{j} + t^2\hat{k}$. (ii) $v(1) = 2\hat{i} + 2\hat{j} + \hat{k}$. $v(1) = \sqrt{4+4+1} = 3$. (iii) $a(t) = v'(t) = 2\hat{i} + 0\hat{j} + 2t\hat{k}$. (iv) $\hat{T} = v(1)/ v(1) = (2\hat{i} + 2\hat{j} + \hat{k})/3$.
4	(i) $\partial\phi/\partial x = 2xyz + z^2$, $\partial\phi/\partial y = x^2z$, $\partial\phi/\partial z = x^2y + 2xz$. At $(1,1,2)$: $\nabla\phi = (2 \cdot 1 \cdot 1 \cdot 2 + 4)\hat{i} + (1 \cdot 2)\hat{j} + (1 + 4)\hat{k} = 8\hat{i} + 2\hat{j} + 5\hat{k}$. (ii) Direction $d = 2\hat{i} - \hat{j} + 2\hat{k}$, $d = 3$, $\hat{a} = (2\hat{i} - \hat{j} + 2\hat{k})/3$. $D_{\hat{a}}\phi = \nabla\phi \cdot \hat{a} = (8 \cdot 2 + 2 \cdot (-1) + 5 \cdot 2)/3 = (16 - 2 + 10)/3 = 24/3 = 8$. (iii) Max rate $= \nabla\phi = \sqrt{64 + 4 + 25} = \sqrt{93}$. Direction: $(8\hat{i} + 2\hat{j} + 5\hat{k})/\sqrt{93}$.
5	(i) $\partial F_1/\partial x = 2xy$, $\partial F_2/\partial y = 2yz$, $\partial F_3/\partial z = 2zx$. $\nabla \cdot F = 2xy + 2yz + 2zx$. At $(1,1,1)$: $\nabla \cdot F = 6$. NOT solenoidal. (ii) $\nabla \times F$: $\hat{i} = [\partial(z^2x)/\partial y - \partial(y^2z)/\partial z] = 0 - y^2 = -y^2$. $\hat{j} = -[\partial(z^2x)/\partial x - \partial(x^2y)/\partial z] = -[z^2 - 0] = -z^2$. $\hat{k} = [\partial(y^2z)/\partial x - \partial(x^2y)/\partial y] = 0 - x^2$. $\nabla \times F = -y^2\hat{i} - z^2\hat{j} - x^2\hat{k}$. At $(1,1,1)$: $\nabla \times F = -\hat{i} - \hat{j} - \hat{k} \neq 0$. NOT irrotational. (iii) F is neither solenoidal nor irrotational at $(1,1,1)$.

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